Bridging Simulation and Experiment with Deep Neural Networks for Estimating Protein-Protein Binding Affinity

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Abstract

Protein–protein binding affinity measures how strongly two proteins associate and is important for protein design, therapeutic targeting, and drug discovery. Most fast predictors estimate affinity from a single bound structure, even though binding is an ensemble property and proteins can adopt more than one relevant arrangement. This study tests whether short explicit-solvent molecular dynamics (MD) can add useful information to a residue-graph affinity predictor.

Starting from a curated wild-type subset of PPB-Affinity, each retained complex was prepared for MD, simulated in OpenMM, converted to a residue graph, and enriched with trajectory-derived summary features. The final controlled comparison used 1437 complexes and 52 models, combining full or interface-restricted graphs, static or static-plus-dynamic features, and two model families: a masked GraphVAE with affinity predicted from exported latent means by ridge regression, and a direct supervised graph regressor. All models were evaluated over 10 outer held-out fits.

The strongest model was the direct supervised baseline on the full static graph, with mean test RMSE 2.3379 ± 0.1067, Pearson correlation 0.5064 ± 0.0567, and $R^2 = 0.2146 \pm 0.0891$. The full graph outperformed the interface-restricted graph in all 24 matched GraphVAE comparisons, and static features outperformed static-plus-dynamic features in 11 of 12 matched GraphVAE comparisons under the static-only reconstruction target.

On this benchmark, the static whole-complex representation was strongest. The results suggest that 1.2 ns trajectories captured little new affinity-relevant motion, while coarse trajectory summaries made any weak added signal difficult to use reliably.

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Chapter 1

Introduction

Protein–protein binding affinity prediction asks how strongly two proteins bind. In benchmark datasets for this task, the target is a thermodynamic quantity such as dissociation constant K_D or binding free energy ΔG, while the input is usually a single three-dimensional complex structure [1]. Accurate affinity prediction is important for protein design, therapeutic targeting, and drug discovery [2, 3]. Experimental measurements provide the ground truth, but they are too slow and expensive for large exploratory searches. Physics-based free-energy calculations aim more directly at binding free energy, but the more faithful versions are also expensive [4, 5]. Many practical machine-learning predictors therefore estimate affinity quickly from structural input alone.

One difficulty is that binding affinity is an ensemble property: it depends on the range of shapes and relative arrangements the proteins sample across an ensemble, beyond any one frozen structure. Proteins can bend, shift, and sometimes slide against each other as they bind [6], so a single bound complex can miss relevant information. Fast structure-based predictors therefore face two linked design choices. Some recent protein–protein models focus tightly on the interface, while others use whole-complex or hybrid representations [1, 7–9]. Protein–ligand work has also asked whether MD can add useful information beyond one bound structure [10, 11]. Residue graphs make it possible to test both questions in the same protein–protein setting: amino-acid residues become nodes, and residue–residue proximity or contact defines edges.

A curated wild-type, that is, non-mutant, subset of PPB-Affinity [1] provides the data. A matched subset of 1437 complexes was taken from a curated table of 2818. Each retained complex went through the same preparation and short-MD procedure before conversion into a post-preparation residue graph with DeepRank2 and enrichment with trajectory-derived summaries. Affinity was then predicted in two ways: a masked GraphVAE learned graph-level latent means, which were passed to ridge regression, and a direct supervised graph regressor predicted affinity directly from the graph. The matched design therefore isolates the effect of short MD summaries against a static baseline.

Four questions guide the analysis. Is a full residue graph better than an interface-

restricted representation? Do MD-derived summaries improve prediction over static graph features alone? Does a masked GraphVAE produce a better predictive representation than a direct supervised graph regressor with the same encoder architecture? How stable are those answers under repeated held-out evaluation?

Chapter 2 places the work in the literature on protein–protein affinity prediction, molecular simulation, and learned structural representations. Chapter 3 describes the methods. Chapter 4 reports the 52-model comparison. Chapter 5 gives the main conclusions. The appendices cover archived approaches, the complete model table, and the repository layout.

Chapter 2

Literature Review

2.1 Protein–protein benchmark

Protein–ligand benchmarking has had a shared structural resource since PDBbind [12], whereas curated protein–protein datasets for absolute affinity remained much smaller for many years [13]. Protein–protein affinity prediction has therefore developed with less data and with more variation between datasets. PPB-Affinity was introduced to improve that situation by building a larger benchmark specifically for direct protein–protein affinity prediction [1]. Reviews of the field repeatedly point to three practical difficulties: limited affinity data, mixed interaction classes, and proteins changing shape as they bind [2, 5]. Protein–protein affinity prediction is hard because it combines limited data, mixed interaction classes, and structural flexibility.

This makes single-structure scoring especially restrictive. In the Kastritis benchmark, only a minority of complexes were close to rigid-body association, and large or moderate interface rearrangements were common [13]. Many protein pairs bind through bending, shifting, or interface reorganisation instead of simple rigid docking. A useful predictor has to cope with both shape changes within a complex and structural differences across complexes.

PPB-Affinity also shows how heterogeneous this prediction task is. Its benchmark model reached Pearson $r = 0.701$ on the full test set, but subgroup performance was weaker for both antibody–antigen and T-cell receptor–peptide major histocompatibility complex (TCR-pMHC) complexes. The authors linked the TCR-pMHC difficulty to strong interface similarity within that class and noted that both subgroups contain a higher proportion of mutant samples than the full dataset [1]. Later antibody-specific models make the same point from another angle: Yang et al. argued that general protein–protein affinity predictors often transfer poorly to antibody–antigen complexes and proposed specialised regressors instead [14]. Subgroup performance is a central issue in this field. Different interaction classes combine different affinity ranges, interface similarities,

and mutation patterns, so they can reward different representations.

2.2 Protein–protein methods

Umbrella sampling and alchemical free-energy methods aim more directly at binding free energy, but they are expensive and sensitive to protocol design [4, 15]. Most of that literature is protein–ligand focused. For protein–protein systems, the same calculations are usually harder because the complexes are larger and more flexible [5]. MM/PBSA and MM/GBSA are cheaper endpoint approximations [16], but they still depend strongly on setup choices and require careful interpretation. At the other end of the spectrum, fast empirical models use compact structural summaries. PRODIGY predicts affinity from interface contact counts and non-interacting surface descriptors [17], while Moal et al. used engineered whole-complex descriptors [7].

Current learned protein–protein models make different choices about how much structure to keep and how tightly to focus on the interface. PRODIGY and the PPB-Affinity benchmark model are explicitly interface-focused [1, 17]. Moal et al. and PPI-Affinity instead build descriptor sets from the complex as a whole rather than from a fixed interface crop [7, 8]. PPI-Graphomer sits between those extremes: it builds an interface-aware module, but the final predictor combines that interface representation with complete sequence and structure information [9]. Representation scope is therefore a central design choice across these models. Interface-focused methods keep a cleaner local signal, while whole-complex and hybrid methods retain more structural context. A matched whole-complex versus interface comparison is useful for that reason.

2.3 Protein–ligand comparison

Protein–ligand affinity prediction differs from protein–protein affinity prediction in ways that matter here. In protein–ligand work, the ligand is usually much smaller, the interaction is more tightly localised to one binding pocket, and larger shared resources such as PDBbind have been available for much longer [12]. Protein–protein systems are larger, more heterogeneous, and often require the model to decide how much of two extended molecular surfaces should be retained [2, 5]. In protein–ligand prediction, pocket-restricted voxel or graph representations are often a natural choice. In protein–protein prediction, the same design can discard context that still contributes to binding strength.

Protein–ligand work suggests two useful lessons. Pafnucy showed that a fixed local 3D representation could support successful affinity prediction when the binding problem is tightly localised [18]. More recent work has also used MD. Dynaformer argues that static structures can miss affinity-relevant information present in conformational ensembles, while Timenucy and Videonucy learn from frame sequences rather than from simple

Domain	Method	Representation / scope	Model class	Reported result
Protein–protein	PRODIGY [8, 17]	Interfacial contacts and non-interacting surface descriptors; interface-only	Empirical regression	Original paper: $R = -0.73$ on a cleaned benchmark in the paper's sign convention; later PPI-Affinity re-evaluation: 79-complex benchmark subset $r = 0.74$, MAE 1.4 kcal mol^{-1}; external 90-complex set $r = 0.31$, MAE 2.5 kcal mol^{-1}
Protein–protein	Moal et al. [7]	Engineered descriptor set over diverse complexes; whole-complex with extra terms for flexibility and rearrangement	Ensemble of regressors	Diverse 137-complex benchmark: $r = 0.73$, RMSE 2.06 kcal mol^{-1}
Protein–protein	PPI-Affinity [8]	Structure-derived descriptor vector; whole-complex descriptors	Ensemble regressor	Vangone–Bonvin benchmark: $r = 0.62$, MAE 1.8 kcal mol^{-1}; external 90-complex hold-out: $r = 0.50$, MAE 1.8 kcal mol^{-1}
Protein–protein	PPB-Affinity benchmark model [1]	128 interface residues and pair geometry; interface-restricted	Geometry-aware deep network	Whole-dataset benchmark: Pearson $r = 0.701$, Spearman $\rho = 0.691$
Protein–protein	PPI-Graphomer [9]	Complete sequence and structure features combined with an interface-aware module; hybrid complete-plus-interface	Graph transformer	Two external test sets: Pearson $r = 0.641$ and 0.625

Table 2.1: Representative reported results from the protein–protein affinity literature. Reported correlations are taken from the original papers; the datasets differ across rows.

trajectory averages [10, 11]. Together these papers suggest that local representations can work well when the interaction is strongly localised, and that dynamic information is more likely to help when the model keeps richer trajectory information than a small set of summary statistics.

2.4 Generalisation and evaluation

Reported accuracy in this area depends strongly on how train and test examples are separated. PPI-Affinity reported that, in its own re-evaluation, PRODIGY reached $r = 0.74$ on a 79-complex benchmark subset but only $r = 0.31$ on a separate 90-complex hold-out set [8]. Moal et al. had already tried to reduce the same problem by removing homologous complexes, so evaluation stayed clear of near-duplicate training examples [7]. In protein–ligand scoring, Graber et al. later showed that train–test leakage between PDBbind and CASF had inflated benchmark performance, and that stricter splits caused clear drops for previously strong models [19]. Split design shapes the scientific interpretation: reported gains can come from a stronger model, an easier split, or both.

2.5 Variational models

Variational autoencoders were introduced as latent-variable models that combine reconstruction with a regularised latent distribution [20]. Variational graph autoencoders brought the same idea to graph data [21], and later GraphVAE-style models made graph generation more explicit [22]. In molecular machine learning, related latent-space VAEs were already being used successfully for small molecules [23]. Here the point is narrower than full graph generation: the GraphVAE is used to learn a compact residue-graph representation and then test whether that representation carries useful affinity signal.

Simulation work also supports this choice. Tian et al. showed that a VAE trained on protein MD trajectories could learn a latent space that captured physically plausible structural variation [24]. That result suggests that a latent model may be useful when structure and MD features are high-dimensional and only part of that information is relevant to binding strength.

2.6 Position of this work

Recent protein–ligand work has pushed further on combining MD with learned structural representations, including models that use ensemble information or ordered trajectory frames instead of a single static complex [10, 11]. Protein–protein affinity prediction now has larger curated resources such as PPB-Affinity, but the problem remains harder because the complexes are larger, the interaction classes are more mixed, and the choice between interface-focused and whole-complex representation is less settled [1, 2, 5]. This thesis brings those threads together by asking whether short explicit-solvent MD summaries help protein–protein affinity prediction, and by comparing that added dynamic information in matched whole-complex and interface-restricted residue-graph settings.

Chapter 3

Methods

3.1 Dataset and target

All experiments used a curated wild-type subset of PPB-Affinity [1]. Wild-type here means non-mutant complexes. For consistency, each complex was stored with one ordered assignment of the two binding partners, labelled ligand and receptor. Curation retained one ordered ligand–receptor chain assignment per complex, removed duplicate entries, and excluded mutant complexes. A complex was kept only if a Protein Data Bank structure, an interpretable ligand/receptor chain definition, and an affinity measurement sufficient to derive a binding free energy were available. The resulting curated table contained 2818 complexes.

The interaction classes in this curated table were 199 T-cell receptor–peptide major histocompatibility complex (TCR-pMHC) complexes, 437 antibody–antigen complexes, and 2182 other or unannotated complexes. PPB-Affinity reports different affinity distributions and weaker benchmark performance for the TCR-pMHC and antibody–antigen subsets than for the full test set [1]. The dataset is therefore heterogeneous, and subgroup composition has to be tracked throughout the study.

The regression target was the standard Gibbs free energy of binding, ΔG, in kcal mol^{-1}. In this curated table, the affinity column was the reported dissociation constant K_D. All targets were therefore computed as

$$\Delta G = RT \ln K_D, \tag{3.1}$$

with $R = 0.00198720425864083$ kcal mol^{-1} K^{-1} and the temperature column `Temp_K`. Where a source temperature was unavailable, the study used 298.15 K.

3.2 Molecular-dynamics preparation and simulation

Complexes were prepared from PDB coordinates with PDBFixer and the OpenMM application layer [25, 26]. The same short-MD procedure was applied to every complex: extract the two partner chains, clean the experimental structure, choose hydrogen and charge states, add water and salt, and then run a short simulation. The annotated ligand and receptor chains were extracted first. Missing residues were detected but not rebuilt. Nonstandard residues were converted when a standard mapping was available, heterogens were removed, missing heavy atoms (that is, non-hydrogen atoms) were added, and hydrogens were assigned before solvation. The same preparation choices were then applied across complexes rather than tuned case by case.

Prepared complexes were solvated with 1.0 nm padding and ionic strength 0.15 M. Protein parameters came from `amber14-all.xml` and water from `amber14/tip3pfb.xml`, corresponding to Amber protein parameters together with the TIP3P-FB water model [27, 28]. No per-complex pH values were available, so protonation, that is, assignment of hydrogen and charge states, was carried out at pH 7.0 for every complex.

All simulations were run in OpenMM [26] with particle mesh Ewald (PME) electrostatics, a 1.0 nm real-space cutoff, constraints on bonds involving hydrogen, rigid water, a Monte Carlo barostat at 1 atm, a Langevin middle integrator with friction 1 ps^{-1}, and a 2 fs timestep. Each system was simulated at the temperature recorded for that complex in `Temp_K`; where no source temperature was available, the same 298.15 K fallback used for the ΔG conversion was used in simulation. Each system was minimised for up to 1000 iterations, equilibrated for 100,000 steps, and then simulated for 500,000 further steps. Coordinate frames were written every 5000 steps. Because recording began before equilibration, each complex contributed 120 sampled frames over 1.2 ns in total.

3.3 Residue-graph construction and features

Residue graphs were built with DeepRank2 [29] from the post-preparation protein topology written after preparation. Both the S and SD modes used the same post-prep graph, so the only extra information in SD came from the added trajectory summaries. A query radius of 10^6 Å ensured that the entire prepared complex was retained. DeepRank2 supplied the basic residue and edge descriptors for residue identity, contact geometry, solvent exposure, and surface area.

The graph inputs were organised in a 2×2 layout: static versus dynamic features, each at node and edge level. Static features were present in both model modes. Dynamic features were added only in the SD mode. Because the underlying post-prep graph is shared, the incremental difference in SD is the added trajectory-derived information. Dynamic information entered as residue- and edge-level summary features on a fixed

graph, with time order left implicit in those summaries.

- **Static node features**
  - **DeepRank2 geometry and exposure descriptors:** buried surface area, solvent-accessible surface area, residue depth, and three half-sphere exposure measures. The half-sphere exposure channels count neighbouring residues in different local directions around each residue.
  - **Neighbour counts:** numbers of nearby residues within 8 Å on the same chain and on the partner chain, computed on the static prepared structure.
  - **Residue identity and physicochemical descriptors:** a 20-way amino-acid one-hot vector, broad residue-class indicators (non-polar, polar, charged), physicochemical scalars (side-chain size, net charge, hydrophobicity, α-helix propensity, β-sheet propensity), and N- and C-terminus flags.
- **Static edge features**
  - **Residue–residue distance:** the static distance attached to each connected residue pair in the prepared structure.
  - **Same-chain indicator:** a binary flag marking whether the two residues belong to the same protein chain.
  - **DeepRank2 interaction terms:** electrostatic and van der Waals edge descriptors supplied by DeepRank2 for each residue pair.
- **Dynamic node features**
  - **Hydration summaries:** computed from the full solvated trajectory with MDTraj [30] by counting unique water oxygen residues within 5 Å of any heavy atom of a residue in each sampled frame. The resulting features were $\log(1+\bar{w}_i)$, where $\bar{w}_i$ is the mean framewise water count for residue i, together with the standard deviation of the framewise counts.
  - **Cα flexibility:** alpha-carbon root-mean-square fluctuation (Cα RMSF), computed after least-squares alignment of the trajectory frames.
  - **Backbone torsion summaries:** frame-averaged values of $\sin\phi$, $\cos\phi$, $\sin\psi$, and $\cos\psi$, where ϕ and ψ are the standard peptide-backbone dihedral angles about the N–Cα and Cα–C bonds.
  - **Inter-chain force summaries:** coarse residue-level interaction proxies from a protein-only OpenMM analysis system rebuilt from the prepared protein topology. The original nonbonded parameters were copied into separate Coulomb

Dataset	n	Other	Antibody–antigen	TCR-pMHC	Mean ΔG
Curated wild-type table	2818	2182 (77.4%)	437 (15.5%)	199 (7.1%)	-9.88 ± 2.57
Matched final subset	1437	966 (67.2%)	366 (25.5%)	105 (7.3%)	-10.12 ± 2.65

Table 3.1: Composition of the full curated wild-type table and the retained 1437-complex matched subset. The retained set is enriched for antibody–antigen complexes and has a slightly more negative mean binding free energy. Percentages and ΔG summary statistics are rounded.

and Lennard–Jones `CustomNonbondedForce` objects [26], restricted to ligand-atom versus receptor-atom interaction groups, accumulated from atom-level forces to residue-level Euclidean norms, and stored as framewise means and standard deviations for the electrostatic and van der Waals channels.

- **Dynamic edge features**
  - **Closest heavy-atom distance summaries:** for each residue pair already connected in the static graph, the closest heavy-atom distance was evaluated in every sampled frame and summarised by its mean and standard deviation.
  - **Dynamic contact frequency:** for the same connected residue pairs, the fraction of sampled frames with closest-heavy distance at most 8 Å.

3.4 Graph views, analysis subset, and feature scaling

Two graph views were evaluated. The *full* view retained the complete residue graph. The *interface-restricted* view first identified the closest inter-partner Cα residue pair in the first protein-only MD frame, then retained up to 128 residues closest to that pair by residue anchor coordinates. The full view keeps the whole complex, whereas the interface view keeps only residues near the binding site. The 128-residue cap follows the fixed interface-window design used in the PPB-Affinity benchmark model [1].

The matched comparison retained 1437 of the 2818 curated complexes. MD simulation, feature extraction, and repeated model evaluation were carried out on this smaller subset. The paired full/interface analysis used complexes with both graph views and the full MD-derived feature set available. The retained set contained 105 TCR-pMHC, 366 antibody–antigen, and 966 other complexes. Table 3.1 compares the retained set with the full curated table.

Feature scaling was fit separately inside each outer split and only on that split's training complexes. For each node or edge feature, all training values were pooled across residues or edges, truncated to the training 0.5th and 99.5th percentiles, centred by the training mean, scaled by the training standard deviation, and finally clipped to the z-score range $[-8, 8]$. The same frozen transformation was then applied to the validation and test complexes.

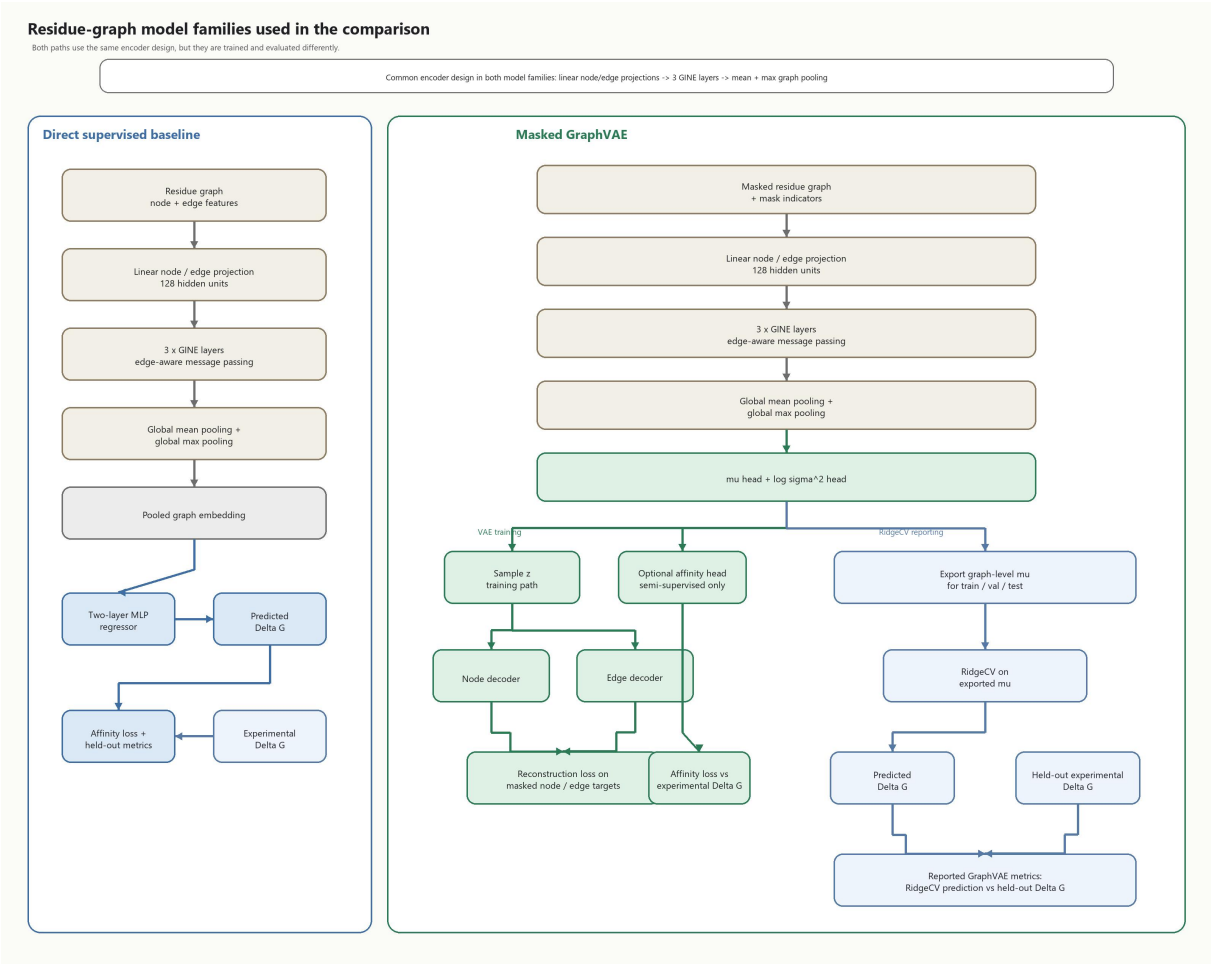

Figure 3.1: Model structure used in the comparison. The two model families use the same encoder architecture: linear node and edge projections, three GINE layers, and mean-plus-max graph pooling. The supervised baseline predicts ΔG directly from the pooled graph embedding. The GraphVAE maps the pooled graph embedding to μ and $\log \sigma^2$, uses sampled latent vectors for masked reconstruction, uses the latent mean for the optional affinity head during training, and reports affinity from ridge regression on latent means.

3.5 Models

Both model families used the same residue-graph encoder architecture. The supervised baseline predicted affinity directly from the pooled graph representation, whereas the GraphVAE learned a masked latent representation and affinity was then predicted from the latent means with ridge regression.

Figure 3.1 shows the shared encoder backbone and the two model paths used in the comparison.

3.5.1 Direct supervised baseline

The supervised baseline was a residue-graph regressor in PyTorch Geometric [31]. Its task is simple: compress each residue graph to one graph-level vector and predict ΔG directly from that vector. Node and edge inputs were projected linearly to 128 hidden units, passed through three graph isomorphism network layers with edge features (GINE), pooled with the concatenation of global mean pooling and global max pooling, and mapped to ΔG by a two-layer multilayer perceptron [32, 33]. Training used mean-squared error loss, AdamW with learning rate 10^{-4} and weight decay 10^{-5}, batch size 16, maximum 120 epochs, and early stopping after 20 epochs without validation improvement. These settings were kept fixed across all four baseline variants.

3.5.2 Masked GraphVAE

The GraphVAE used the same encoder architecture as the supervised baseline: linear node and edge projections to 128 hidden units, three GINE layers, and mean-plus-max graph pooling. Two linear heads mapped the pooled representation to the latent mean μ and log-variance $\log \sigma^2$. The latent dimensionality was one of $d_z \in \{8, 16, 32\}$. The masking task trains the model to fill in deliberately hidden feature values from the rest of the graph. The goal is to learn a compact graph representation for affinity prediction.

For every batch, target node entries and target edge entries were masked independently with probability 0.30. The encoder received the masked graph together with binary mask indicators. The node decoder received the graph latent, the unmasked non-target node inputs, the masked target node inputs, and the node-mask indicators. The edge decoder received the analogous edge quantities. The reconstruction task was defined feature by feature for selected node and edge channels, instead of asking the model to generate an entire graph from scratch [21, 22].

The reconstruction targets used in the 52-model comparison were:

- **Static-only target:** buried surface area, solvent-accessible surface area, residue depth, the three half-sphere exposure channels, same-chain and partner-chain neighbour counts within 8 Å, and edge distance.
- **Dynamic-only target:** all dynamic node features and all dynamic edge features.
- **Static-plus-dynamic target:** the union of the static and dynamic target sets.

Residue-identity indicators and the same-chain edge flag were input features only; they were excluded from the reconstruction targets.

For a batch of graphs, the GraphVAE objective was

$$\mathcal{L} = \mathcal{L}_{\text{node}} + \mathcal{L}_{\text{edge}} + \beta \mathcal{L}_{\text{KL}} + 0.01\, \mathcal{L}_{\text{corr}} + \lambda_{\text{aff}} \mathcal{L}_{\text{aff}}, \tag{3.2}$$

where β is the weight on the Kullback–Leibler regulariser, $\mathcal{L}_{\text{node}}$ and $\mathcal{L}_{\text{edge}}$ were masked mean-squared reconstruction losses, $\mathcal{L}_{\text{corr}}$ penalised the squared off-diagonal entries of the latent covariance matrix to discourage redundant latent coordinates, and $\mathcal{L}_{\text{aff}}$ was an MSE loss on normalised ΔG from a latent MLP head. Unsupervised runs used $\lambda_{\text{aff}} = 0$. Semi-supervised runs used $\lambda_{\text{aff}} = 1$. The Kullback–Leibler term was

$$\mathcal{L}_{\text{KL}} = \frac{1}{B} \sum_{b=1}^{B} \frac{1}{d_z} \sum_{j=1}^{d_z} \left[-\frac{1}{2} \left(1 + \log \sigma_{bj}^2 - \mu_{bj}^2 - \sigma_{bj}^2 \right) \right], \tag{3.3}$$

where μ_{bj} is the latent mean for graph b and latent coordinate j, and σ_{bj}^2 is the corresponding latent variance. The model predicts μ and $\log \sigma^2$, so σ^2 is obtained by exponentiating the predicted log-variance. Here B is the batch size and d_z is the latent dimensionality, so the loss averages the Kullback–Leibler penalty over graphs and latent coordinates.

All GraphVAE models were trained with AdamW, learning rate 10^{-4}, weight decay 10^{-5}, batch size 16, maximum 120 epochs, and early-stopping patience 20. Gradient norms were clipped at 5 during optimisation. The KL weight increased linearly from 0 to 1 over the first 30% of training epochs, following the common deterministic warm-up strategy for variational training [34]. Checkpoint selection used validation reconstruction loss for unsupervised runs and validation reconstruction loss plus weighted validation affinity loss for semi-supervised runs.

3.5.3 Latent-space ridge regression

GraphVAE results are reported from a linear ridge regressor fit to the latent means. The neural affinity head was used during semi-supervised training, but the reported GraphVAE numbers come from one shared linear downstream model so that the comparison stays focused on the information already present in the learned representation. After training, one graph-level latent-mean vector was extracted per complex for the train, validation, and test splits.

For training complexes with latent matrix $\mathbf{Z}_{\text{train}}$ and target vector $\mathbf{y}_{\text{train}}$, the ridge model solves

$$\hat{\mathbf{w}} = \arg\min_{\mathbf{w}} \|\mathbf{y}_{\text{train}} - \mathbf{Z}_{\text{train}}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_2^2, \tag{3.4}$$

where α controls the strength of ℓ_2 regularisation. The regularisation strength was selected by 5-fold inner cross-validation on the training latents with `RidgeCV` from scikit-learn [35], searching the grid $\{10^{-4}, 3 \times 10^{-4}, 10^{-3}, 3 \times 10^{-3}, 10^{-2}, 3 \times 10^{-2}, 10^{-1}, 3 \times 10^{-1}, 1, 3, 10, 30, 100\}$. The resulting fitted linear model was then applied without refitting to the train, validation, and test complexes in that outer split. The headline GraphVAE metrics in Chapter 4 are the held-out metrics of this ridge regression.

3.6 Repeated outer evaluation

The 52-model comparison was evaluated on the matched 1437-complex set with 5 outer folds and 2 repeats. Repeated outer resampling was used because a single split can give a misleading picture of comparative performance when many closely related models are tested [36]. Resampling operated at the level of complexes, so all trajectory frames from a complex stayed in the same split. In each outer fit, one fold was held out as test and the remaining four folds were split into training and validation sets with validation fraction 0.1875 of the training-plus-validation pool, giving an overall train/validation/test proportion of approximately 65/15/20. At each outer fit, feature scaling, model training, checkpoint selection, and latent ridge fitting were all carried out within that split. Split generation and model fitting used fixed pseudorandom seeds derived deterministically from the outer repeat and fold.

The comparison contained 52 models. Forty-eight were GraphVAE models spanning two graph views (full and interface-restricted), two feature modes (S and SD), two supervision modes (unsupervised and semi-supervised), and latent dimensions 8, 16, and 32. The SD runs used three reconstruction targets; the S runs used only the static-only target. Four further models were direct supervised baselines for the Cartesian product of graph view and feature mode. Throughout the results chapter, S denotes static-only inputs, SD denotes static-plus-dynamic inputs, and d_z or z denotes latent dimension.

Held-out performance was measured with RMSE, Pearson correlation, and R^2. Every metric reported in Chapter 4 is the arithmetic mean over the 10 outer fits for the corresponding model, and the corresponding standard deviations are reported from the same ten fits.

Chapter 4

Results

The final evaluation used the 1437-complex analysis subset for which paired full and interface-restricted graph views were available. The comparison contained 52 models: 48 GraphVAE variants and 4 direct supervised baselines. Every reported metric is the mean over 10 outer held-out fits from 5 folds and 2 repeats. Here S denotes static-only inputs, SD static-plus-dynamic inputs, *semi* semi-supervised GraphVAE training, *unsup* unsupervised GraphVAE training, and z latent dimension. The complete 52-model table is given in Appendix B.

4.1 Headline models

The full static supervised baseline ranked first overall. The best GraphVAE was the full static semi-supervised model with latent dimension 32, and the best interface model was the static supervised baseline. Figure 4.1 and Table 4.1 list the ranking and headline values.

The best supervised baseline outperformed the best GraphVAE by 0.0913 RMSE units and 0.0415 Pearson-correlation units on the mean metrics. Across the 10 outer fits, it had lower test RMSE in 8 of 10 splits than the best GraphVAE.

4.2 Factor comparisons

Graph view had the strongest effect: the full graph outperformed the interface-restricted graph in all 24 matched GraphVAE comparisons. Under the static-only reconstruction target, the static representation outperformed the static-plus-dynamic representation in 11 of 12 matched GraphVAE comparisons. Semi-supervision helped consistently for the full-graph runs and the interface static runs. For the interface dynamic runs, its effect was mixed: mean correlation improved, but mean RMSE was slightly worse. This is consistent with the family-level train/validation/test behaviour: in several semi-supervised

Model	Test RMSE	Test r	Test R^2	Test−Train RMSE
Full baseline, S	2.3379 ± 0.1067	0.5064 ± 0.0567	0.2146 ± 0.0891	0.8021
Full baseline, SD	2.4073 ± 0.1290	0.4802 ± 0.0494	0.1685 ± 0.0876	0.8414
Full GraphVAE, S, semi, $z = 32$	2.4292 ± 0.0809	0.4649 ± 0.0363	0.1537 ± 0.0674	0.7451
Full GraphVAE, SD, semi, static-only, $z = 8$	2.4364 ± 0.1065	0.4749 ± 0.0529	0.1478 ± 0.0860	1.1303
Interface baseline, S	2.5782 ± 0.1120	0.3451 ± 0.0572	0.0470 ± 0.0820	0.8090
Interface GraphVAE, S, semi, $z = 8$	2.6041 ± 0.1019	0.3402 ± 0.0611	0.0282 ± 0.0718	0.8018

Table 4.1: Best-performing models in the 52-model comparison. Reported RMSE, Pearson r, and R^2 values are mean ± standard deviation over the 10 outer fits. GraphVAE metrics are those obtained after ridge regression on latent means.

Comparison	Pairs	Mean RMSE gain for first option	Mean Pearson r gain for first option	Pair wins for first option
Full over interface	24	0.1005	0.0993	24 / 24
S over SD (shared static)	12	0.0254	0.0215	11 / 12
Semi over unsup, full S	3	0.1640	0.2497	3 / 3
Semi over unsup, full SD	9	0.1604	0.3339	9 / 9
Semi over unsup, interface S	3	0.0290	0.2436	3 / 3
Semi over unsup, interface SD	9	-0.0081	0.2766	3 / 9

Table 4.2: Matched GraphVAE comparisons. A positive RMSE gain means the first-named option has lower mean test RMSE.

families, SD models fit training data more strongly while still giving weaker held-out performance. Table 4.2 lists the numerical summary, and Figure 4.1 shows the same comparisons visually.

Within the full semi-supervised SD family, the reconstruction target also changed performance. Averaged over latent dimensions 8, 16, and 32, the mean test RMSE was 2.4434 for the static-only target, 2.4795 for the static-plus-dynamic target, and 2.5150 for the dynamic-only target. The corresponding mean test Pearson correlations were 0.4682, 0.4335, and 0.4221.

4.3 Stability under repeated evaluation

The main qualitative conclusions stayed stable across repeated evaluation, although the margin between the two best full-graph models was modest. The full static baseline achieved mean test RMSE 2.3379 ± 0.1067, while the best GraphVAE achieved 2.4292 ± 0.0809. Across the 10 outer fits, the full static baseline had lower test RMSE than the best GraphVAE in 8 of 10 splits and lower test RMSE than the interface static baseline in 9 of 10 splits. The full-over-interface effect was even clearer at the design level, because the full graph won all 24 matched GraphVAE comparisons.

The same ranking and the same directional factor effects recurred across multiple train/test partitions, even though the top two full-graph models stay close on a few splits.

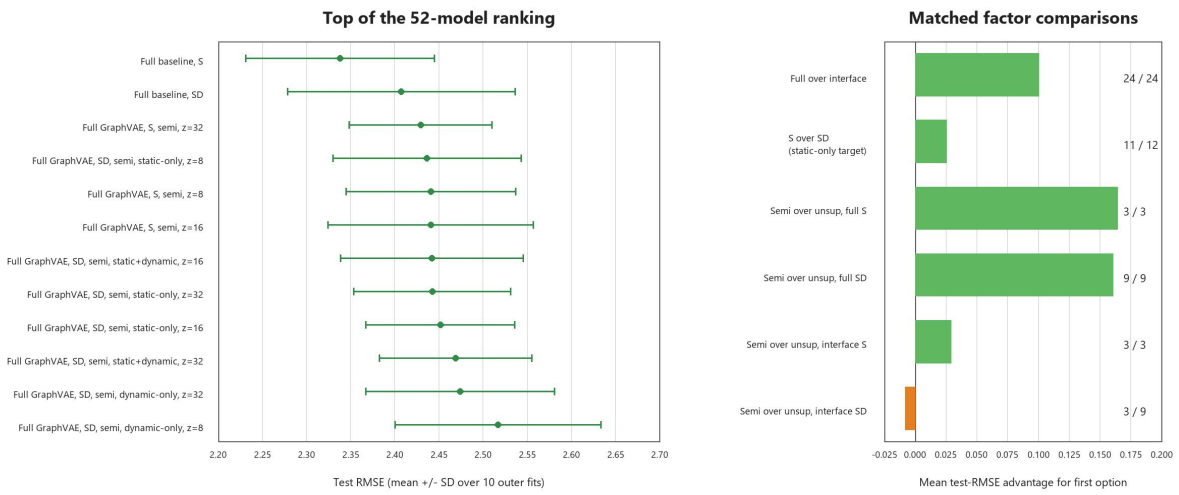

Figure 4.1: Main empirical picture of the 52-model comparison. Left: top of the model ranking by mean test RMSE with one-standard-deviation error bars over the 10 outer fits. Right: matched GraphVAE factor comparisons, reported as the mean test-RMSE advantage for the first-named option.

4.4 Generalisation by model family

The family-level averages separate in-sample fit from held-out prediction. In the semi-supervised families, adding dynamic summaries often increased train correlation and reduced train RMSE, yet those gains faded on validation and test. The unsupervised VAE families had small train–test gaps but weak correlations on all splits, which is more consistent with underfitting than with strong regularised prediction. Across all families, the strongest held-out result remained the full static supervised baseline. Table 4.3 lists the corresponding family means.

The clearest example is the full semi-supervised GraphVAE family: moving from S to SD raised mean train correlation from 0.7494 to 0.8218 and lowered mean train RMSE from 1.7385 to 1.4744, yet mean test correlation fell from 0.4551 to 0.4413 and mean test RMSE worsened from 2.4369 to 2.4793. The same pattern appears in the interface semi-supervised families. The dynamic summaries gave the models more variation to fit, but that extra fit stayed on the training set instead of improving held-out prediction.

4.5 Performance by interaction class

The matched 1437-complex analysis set contained 966 complexes in the broad *other* class, 366 antibody–antigen complexes, and 105 TCR-pMHC complexes. For subgroup analysis, the outer-test predictions were pooled over all 10 fits, so each complex contributes two held-out predictions, one from each repeat.

Subgroup analysis told the same story. The full static baseline remained the strongest

Family	Train r	Val r	Test r	g_r	Train RMSE	Test RMSE
Full baseline S	0.8225	0.5337	0.5064	0.3161	1.5358	2.3379
Full baseline SD	0.8077	0.5191	0.4802	0.3275	1.5659	2.4073
Full VAE unsup S	0.2486	0.1897	0.2055	0.0431	2.5734	2.6009
Full VAE unsup SD	0.1809	0.0967	0.1074	0.0736	2.6062	2.6398
Full VAE semi S	0.7494	0.4666	0.4551	0.2942	1.7385	2.4369
Full VAE semi SD	0.8218	0.4580	0.4413	0.3805	1.4744	2.4793
Interface VAE unsup S	0.1495	0.0794	0.0925	0.0570	2.6289	2.6402
Interface VAE unsup SD	0.1261	0.0382	0.0424	0.0838	2.6324	2.6538
Interface VAE semi S	0.7083	0.3356	0.3361	0.3722	1.8467	2.6112
Interface VAE semi SD	0.7685	0.3231	0.3189	0.4496	1.6479	2.6620

Table 4.3: Family-level mean train, validation, and test behaviour. Here $g_r = r_{\text{train}} - r_{\text{test}}$. Each row is averaged over all models in that family.

Subgroup	n	Full baseline RMSE	Full baseline r	Full GraphVAE RMSE	Full GraphVAE r	Interface baseline RMSE	Interface baseline r
Antibody–antigen	366	2.048	0.292	2.134	0.193	2.356	0.089
Other	966	2.501	0.430	2.584	0.394	2.675	0.312
TCR-pMHC	105	1.663	0.466	1.879	0.406	2.440	0.333

Table 4.4: Pooled outer-test subgroup metrics for the best overall model (full static baseline), the best GraphVAE (full static semi-supervised GraphVAE, latent dimension 32), and the best interface model (interface static baseline).

of these models in all three interaction classes. The full static GraphVAE remained second, and the interface baseline remained weaker in every subgroup. The weakest correlations were in the antibody–antigen subgroup, while the TCR-pMHC subgroup had the smallest sample size and therefore the widest uncertainty. Table 4.4 and Figure 4.2 give the subgroup values and outer-fit variability.

4.6 Summary

Four results dominate the comparison. First, the full graph was better than the interface-restricted representation. Second, the dynamic summaries offered no held-out gain over the static representation, even though they sometimes improved in-sample fit. Third, the strongest model was the direct supervised baseline on the full static graph. Fourth, the same model ordering remained in the antibody–antigen, TCR-pMHC, and other interaction classes, and the repeated-resampling analysis showed that these conclusions hold across multiple held-out splits.

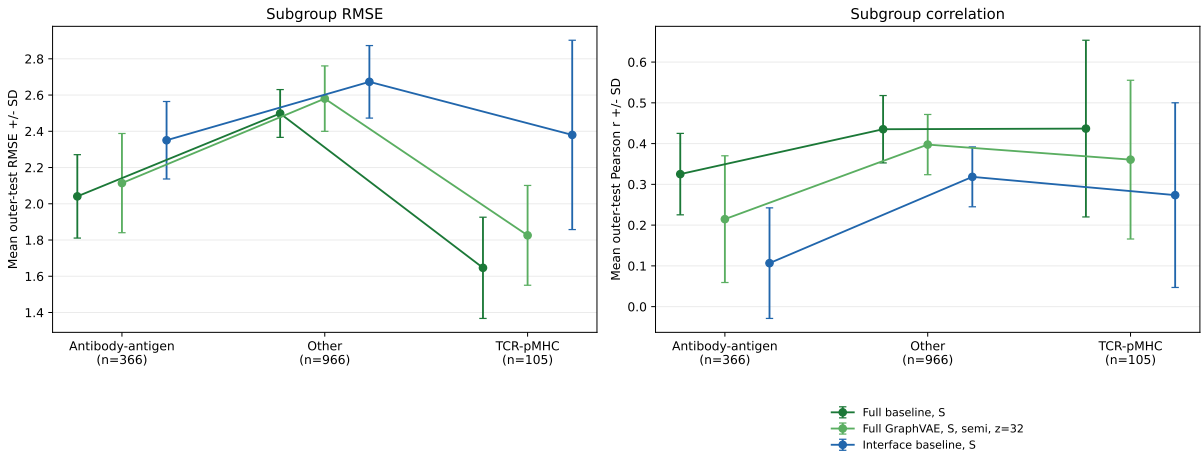

Figure 4.2: Subgroup performance of the three headline models. Points show the mean outer-test metric and error bars show one standard deviation across the 10 outer fits. The model ordering is stable across subgroups, but uncertainty is visibly larger for the 105-complex TCR-pMHC subset.

Chapter 5

Discussion and Conclusions

5.1 Main findings

First, the full residue graph outperformed the interface-restricted representation. The full view won all 24 matched GraphVAE comparisons and both supervised baseline comparisons. Whole-complex context appears to help the model recognise which partners are present, how the two chains are arranged in space, and what broader structural setting surrounds the binding site. A fixed 128-residue interface window keeps only part of that information. On this benchmark, the extra context was more useful than a tighter focus on the contact region. The pattern is closer to whole-complex or hybrid approaches such as Moal et al., PPI-Affinity, and PPI-Graphomer than to strongly interface-focused predictors such as PRODIGY and the PPB-Affinity benchmark model [1, 7–9, 17]. Under the same encoder family and repeated held-out evaluation, the whole-complex view was the stronger choice.

Second, the static feature set remained stronger than the MD-derived summaries. The comparison between S and SD was intentionally controlled: both modes started from the same post-prep graph, so the only extra information in SD came from the trajectory summaries. Within that shared setup, the family-level results show the same pattern in both the full and interface semi-supervised families. Adding dynamic summaries improved train fit but weakened validation and test performance. In practice, the added MD summaries increased in-sample flexibility without improving held-out prediction.

Third, the masked GraphVAE remained competitive, and semi-supervised training improved most of the VAE families over the unsupervised variants. The exception was the interface dynamic family, where the effect was mixed. The full static baseline still produced the strongest held-out predictor. A likely explanation is that direct supervised training can spend all of its capacity on affinity, whereas the VAE also spends capacity reconstructing masked graph features. That trade-off still produced useful latent spaces, but direct supervised training was more effective for affinity prediction in this comparison

[20, 21, 24].

Fourth, the main conclusions were stable under repeated held-out evaluation. The same design choices stayed favourable across 10 outer fits, and the family-level summaries support the same ranking as the leading models. The full static baseline beat the best GraphVAE in 8 of 10 outer fits, so the top of the ranking holds even though the two best full-graph models remain fairly close.

Interaction classes also differed meaningfully. On pooled outer-test predictions, the best overall model, the full static baseline, gave Pearson $r = 0.292$ on antibody–antigen, $r = 0.430$ on the broad other class, and $r = 0.466$ on TCR-pMHC. The full static baseline remained the strongest of the main models in all three interaction classes. This matches PPB-Affinity's subgroup behaviour and later antibody-specific work arguing that antibody–antigen complexes benefit from specialised treatment [1, 14]. A pooled benchmark average can hide those differences: a model can look competitive overall and still struggle on an interaction class that matters in practice. The antibody–antigen correlations here are clearly weaker than the broad other class, while the TCR-pMHC subgroup is small enough that its uncertainty remains large.

5.2 Limitations

Trajectory length was the main limitation. Each complex contributed 1.2 ns of saved MD, which is enough to capture local fluctuation and contact persistence but is short relative to many rearrangements that shape protein binding [5, 6, 13]. Slow collective motions often extend to microseconds and slower [6], so these trajectories were unlikely to sample much of the slower motion that distinguishes one binding ensemble from another. That makes limited added signal the most likely explanation for the weak dynamic-feature results.

The dynamic features also carry noise from how the MD was run and summarised. With more time and compute, preparation could be refined case by case and the trajectories could be kept in a richer form. Doing so would have reduced the number of complexes that could be carried through the comparison. To keep the study large enough for machine learning, preparation was standardised across complexes, missing residues were not rebuilt, protonation was standardised, and each short trajectory was compressed to residue- and edge-level summary statistics on a fixed graph. Those choices made a 1437-complex comparison feasible, but they also combine thermal fluctuation, finite sampling, and summary-statistic noise. Using the same post-prep graph for both S and SD reduced noise in the comparison itself, because both modes started from the same structure source. That same choice may also leave the absolute static-only numbers below what graphs built directly from raw experimental coordinates could achieve.

The final controlled comparison also used a 1437-complex matched subset drawn from the 2818-complex curated table. That retained set is enriched for antibody–antigen com-

plexes and slightly stronger binders, so the reported numbers should be read as results on this matched subset rather than as a direct replacement for the full PPB-Affinity benchmark. The repeated resampling was done by complex identifier only, leaving homology-aware and structure-aware separation unconstrained, so related protein families could still appear across train and test. Earlier protein–protein studies removed homologous complexes for this reason, and similar leakage problems have been documented in protein–ligand affinity prediction [7, 19]. Transfer to unbound or predicted inputs remains open, since the present study uses bound complex structures in which both proteins are already observed together [2].

5.3 Future work

A stronger test of dynamics would use longer or better-targeted trajectories, especially if the goal is to capture slower binding-related motion as well as local fluctuation. Cleaner structure preparation would also help where the compute budget allows it. Recent protein–ligand work suggests that MD becomes more useful when the model keeps richer temporal or ensemble information instead of reducing the trajectory to a few averages [10, 11].

Richer structure and evaluation settings would also be useful. One shared post-prep graph was used for both S and SD so that the incremental effect of the trajectory summaries stayed isolated. Comparing that controlled design with graphs built directly from raw experimental coordinates would show how much the shared post-prep source affects absolute static performance. Stricter family-aware splits and the same representation choices on unbound or predicted structures would also be useful, since practical affinity prediction is often harder in those settings.

5.4 Conclusion

On this benchmark, a whole-complex static residue graph was the strongest starting point for protein–protein affinity prediction. In the 52-model comparison, representation scope mattered more than the short MD summaries added here: the full static baseline beat every other model, while the dynamic variants more often improved training fit than held-out prediction. The practical lesson is to begin with strong whole-complex static models and careful repeated evaluation, then add dynamics only when longer simulations or models that retain temporal structure can justify the extra complexity.

Appendix A

Archived Approaches

This appendix summarises the main alternative approaches explored before the residue-graph study reported in the main chapters. The archived materials are preserved under `ARCHIVE` in the repository and can also be browsed at `https://github.com/RohanDevereux/Bridging/tree/main/ARCHIVE`.

A.1 Correction learning around fast affinity baselines

An early idea was to predict a correction

$$\delta = \Delta G_{\text{exp}} - \Delta G_{\text{base}}$$

instead of predicting ΔG directly. Two approaches were explored in this form: a contact-based baseline using PRODIGY [17] and an endpoint free-energy baseline using MM/GBSA [16].

A.1.1 PRODIGY

One approach used PRODIGY estimates as a fast structural baseline and then learned a correction to experimental affinity [17]. The corresponding materials are preserved under `ARCHIVE/PRODIGY`. In the saved runs, raw PRODIGY predictions were weak on the archived TCR-heavy subsets, and direct prediction from learned structural representations was stronger than learning a correction around PRODIGY. The archived subgroup summaries also already hinted that the interaction classes behave differently.

The saved subset-specific baseline outputs showed the same pattern. On an archived TCR-only dataset, PRODIGY was essentially uncorrelated with experiment. On a small antibody–antigen archived dataset, the same baseline was better, though still modest. A mixed 100-complex subset was again weak.

The next table switches from raw baseline evaluation to the correction-learning setup.

Archived subset	n	RMSE	MAE	Pearson r	Spearman ρ
TCR-pMHC	65	9.19	7.12	−0.051	−0.044
Antibody–antigen	70	2.33	1.80	0.243	0.236
Diverse100 mixed set	100	6.75	4.66	−0.175	−0.086

Table A.1: Saved subset-specific PRODIGY results from archived curated datasets. The TCR-pMHC subset was especially weak, while the small antibody–antigen subset was somewhat better.

Archived regression variant	PRODIGY n	PRODIGY RMSE	PRODIGY r	Direct RMSE	Direct r	Corrected RMSE
Early contact-map latent regressor	13	10.50	−0.268	6.92	−0.018	8.64
Set-based edge encoder, 24 epochs	13	10.50	−0.268	2.80	−0.500	7.75

Table A.2: Representative archived PRODIGY-based regression runs on the TCR subset. The direct model is the learned latent-space regressor, while the corrected model tries to learn an adjustment to PRODIGY.

It keeps the same PRODIGY baseline column and compares two learned routes on overlapping archived TCR test slices: direct prediction from the learned representation, and prediction of a correction around PRODIGY.

A.1.2 MM/GBSA

A second correction-learning approach used MM/GBSA endpoint estimates [16]. The corresponding materials are preserved under `ARCHIVE/MMGBSA`. Across the partial result files there were 163 rows with status `ok`. On those comparable rows, the MM/GBSA estimates had Pearson correlation −0.127, Spearman correlation −0.096, RMSE 2458.69, and MAE 1398.31. The scale of the predictions was far from the experimental target range: the median successful estimate was 466.38 kcal mol^{-1}, the 95th percentile was 5590.71 kcal mol^{-1}, and the maximum exceeded 13 000 kcal mol^{-1}, whereas the corresponding experimental free energies lay roughly between −21 and −4 kcal mol^{-1}.

A.2 The Cα contact-map approach

An earlier model family represented complexes as fixed two-dimensional tensors built from Cα–Cα contact and distance channels, followed by a convolutional variational autoencoder and regression model. The corresponding materials are preserved under `ARCHIVE/CA_contact_map`. This image-like representation was appealing because related ideas have worked in protein–ligand learning [18], but in the archived runs it was too unstable for the present task. Some variants reduced RMSE relative to raw PRODIGY on small subsets, yet the same runs often had weak or negative test correlations and varied sharply across seeds. The reported values below are test-set outputs from the direct-latent regression path built on that contact-map representation.

Quantity	Value
Successful rows	163
Comparable successful rows	163
Pearson correlation	−0.127
Spearman correlation	−0.096
RMSE	2458.69
MAE	1398.31
Median predicted MM/GBSA ΔG	466.38
95th percentile predicted MM/GBSA ΔG	5590.71
Maximum predicted MM/GBSA ΔG	13628.15

Table A.3: MM/GBSA partial-result summary from the archived approach. The estimates were far outside the scale of experimental protein–protein binding free energies, so this route was dropped as a quantitative baseline.

Run	Test n	Direct-latent RMSE	Direct-latent Pearson r
`setedge_cpu`	8	3.03	−0.065
`setedge_e24`	12	2.80	−0.500
`setedge_e24_ols`	12	2.76	−0.527
`setedge_e24_seed1`	12	6.62	−0.124
`setedge_e24_seed4`	12	3.17	0.462
`setedge_semisup`	12	5.48	0.072
`allmd_setedge_e24`	12	4.23	0.087

Table A.4: Representative Cα contact-map regression runs on the archived TCR subset. Some runs reduced RMSE relative to raw PRODIGY, but the results were unstable and often had weak or negative test correlation.

All runs in Table A.4 share the same basic contact-map representation and the same set-based edge encoder. The labels mainly distinguish training and downstream-regression variants: `e24` marks the 24-epoch latent-model checkpoint used for feature extraction, `seed1` and `seed4` change the random seed, `semisup` adds an auxiliary affinity loss during latent-model training, `ols` records an archived unregularised regression variant, and `allmd` marks a broader MD-frame variant preserved in the archived results. The spread across those runs shows why this branch was left behind.

Compared with the final residue-graph formulation, the contact-map representation threw away residue identity detail, side-chain context, and explicit graph structure while still depending on a fixed interface crop and a fixed tensor ordering. The residue-graph formulation retained more structural information and connected more naturally to the MD-derived features used in the main chapters, so the work later shifted away from the contact-map path.

A.3 PPB-Affinity interface-attention augmentation

Another line of work kept more of the trajectory itself, with baseline, frame-level, and trajectory-level models together with an ensemble stage. The corresponding materials

are preserved under `ARCHIVE/PPB_Affinity_augmentation`. It reused a vendored PPB-Affinity-style baseline stack, including the same kind of geometry-aware attention model over an interface patch, and then changed the data path around it. Here, augmentation meant treating saved MD frames from one complex as additional supervised inputs that all inherited the same affinity label. The branch then explored three linked routes on top of that interface-attention model [1]: a single-structure baseline, frame-level models trained on those augmented examples, and trajectory-level models that passed ordered frame embeddings through a bidirectional gated recurrent unit (BiGRU) before final regression [37, 38]. Recent protein–ligand work had started to learn from MD frame sequences rather than from simple averages [11], so this branch tested whether the same idea could transfer to protein–protein affinity prediction.

The branch combined four ideas. First, the single-structure baseline kept one structure per complex and passed it through the PPB-Affinity-style interface-attention model as a reference path. Second, frame augmentation reused saved MD frames as additional supervised inputs, each with the parent complex's affinity label. Third, the trajectory model kept the frame order and passed frame embeddings through a BiGRU before regression instead of collapsing the trajectory to simple averages. Fourth, an ensemble stage was intended to combine predictions from the baseline, frame-level, and trajectory-level routes.

This branch remained exploratory after the work shifted toward the residue-graph comparison used in the main chapters. That shift says little about the augmentation idea itself. It remains relevant as a follow-on direction because it preserves temporal order that the final summary-feature approach discards.

Appendix B

Complete Model Matrix

This appendix gives the complete 52-model evaluation matrix, sorted by mean test RMSE. A machine-readable copy is stored in the results directory as `final_resample_matrix_summary.csv`. Here *Sup.* denotes supervision status, *Gap* denotes the mean test-minus-train RMSE gap, and the machine-readable label *shared static* denotes the static-only reconstruction target used in the main text.

Table B.1: Complete 52-model evaluation matrix, sorted by mean test RMSE. Metrics are means over 10 outer held-out fits.

View	Model	Mode	Sup.	Target	z	RMSE	r	R^2	Gap
Full	Baseline	S	sup	–	–	2.3379	0.5064	0.2146	0.8021
Full	Baseline	SD	sup	–	–	2.4073	0.4802	0.1685	0.8414
Full	GraphVAE	S	semi	shared static	32	2.4292	0.4649	0.1537	0.7451
Full	GraphVAE	SD	semi	shared static	8	2.4364	0.4749	0.1478	1.1303
Full	GraphVAE	S	semi	shared static	8	2.4407	0.4507	0.1451	0.6921
Full	GraphVAE	S	semi	shared static	16	2.4407	0.4499	0.1438	0.6580
Full	GraphVAE	SD	semi	static+dynamic	16	2.4419	0.4503	0.1437	0.8952
Full	GraphVAE	SD	semi	shared static	32	2.4423	0.4669	0.1443	1.2169
Full	GraphVAE	SD	semi	shared static	16	2.4516	0.4629	0.1354	1.1312
Full	GraphVAE	SD	semi	static+dynamic	32	2.4687	0.4411	0.1233	0.8527
Full	GraphVAE	SD	semi	dynamic only	32	2.4740	0.4508	0.1231	1.0153
Full	GraphVAE	SD	semi	dynamic only	8	2.5168	0.4202	0.0920	0.9580
Full	GraphVAE	SD	semi	static+dynamic	8	2.5277	0.4092	0.0818	0.7819
Full	GraphVAE	SD	semi	dynamic only	16	2.5543	0.3954	0.0629	1.0628
Interface	Baseline	S	sup	–	–	2.5782	0.3451	0.0470	0.8090
Interface	Baseline	SD	sup	–	–	2.5829	0.3347	0.0463	0.6954
Full	GraphVAE	S	unsup	shared static	32	2.5906	0.2271	0.0401	0.0617
Interface	GraphVAE	S	semi	shared static	8	2.6041	0.3402	0.0282	0.8018
Full	GraphVAE	S	unsup	shared static	16	2.6052	0.1910	0.0303	0.0194
Full	GraphVAE	S	unsup	shared static	8	2.6069	0.1984	0.0286	0.0016
Interface	GraphVAE	S	semi	shared static	32	2.6113	0.3364	0.0207	0.7354
Interface	GraphVAE	S	semi	shared static	16	2.6180	0.3316	0.0166	0.7561
Interface	GraphVAE	SD	semi	static+dynamic	16	2.6238	0.3252	0.0146	0.7597
Full	GraphVAE	SD	unsup	shared static	16	2.6313	0.1259	0.0102	0.0265
Full	GraphVAE	SD	unsup	shared static	8	2.6322	0.1249	0.0103	0.0124
Interface	GraphVAE	S	unsup	shared static	32	2.6324	0.1211	0.0102	0.0118
Full	GraphVAE	SD	unsup	static+dynamic	32	2.6328	0.1255	0.0093	0.0610
Full	GraphVAE	SD	unsup	dynamic only	32	2.6353	0.1100	0.0074	0.0364
Full	GraphVAE	SD	unsup	shared static	32	2.6365	0.1813	0.0039	0.1021
Interface	GraphVAE	SD	semi	dynamic only	16	2.6366	0.3298	0.0032	0.9460
Interface	GraphVAE	S	unsup	shared static	8	2.6427	0.0790	0.0025	0.0062
Full	GraphVAE	SD	unsup	dynamic only	16	2.6441	0.0981	0.0013	0.0292
Interface	GraphVAE	S	unsup	shared static	16	2.6454	0.0774	0.0003	0.0157
Full	GraphVAE	SD	unsup	static+dynamic	8	2.6454	0.0719	0.0006	0.0048
Interface	GraphVAE	SD	semi	shared static	16	2.6455	0.3362	-0.0015	1.1836

View	Model	Mode	Sup.	Target	z	RMSE	r	R^2	Gap
Full	GraphVAE	SD	unsup	dynamic only	8	2.6461	0.0706	0.0000	0.0013
Interface	GraphVAE	SD	unsup	shared static	8	2.6473	0.0618	-0.0010	0.0129
Interface	GraphVAE	SD	unsup	static+dynamic	8	2.6481	0.0552	-0.0016	0.0056
Interface	GraphVAE	SD	unsup	shared static	32	2.6530	0.0896	-0.0051	0.0451
Interface	GraphVAE	SD	unsup	static+dynamic	32	2.6534	0.0268	-0.0056	0.0222
Full	GraphVAE	SD	unsup	static+dynamic	16	2.6542	0.0582	-0.0060	0.0286
Interface	GraphVAE	SD	unsup	dynamic only	8	2.6552	0.0293	-0.0071	0.0097
Interface	GraphVAE	SD	unsup	dynamic only	16	2.6554	0.0235	-0.0070	0.0151
Interface	GraphVAE	SD	semi	dynamic only	8	2.6557	0.3279	-0.0116	1.0207
Interface	GraphVAE	SD	unsup	static+dynamic	16	2.6557	0.0209	-0.0073	0.0127
Interface	GraphVAE	SD	unsup	shared static	16	2.6575	0.0395	-0.0088	0.0353
Interface	GraphVAE	SD	unsup	dynamic only	32	2.6589	0.0347	-0.0099	0.0347
Interface	GraphVAE	SD	semi	shared static	8	2.6684	0.3270	-0.0217	1.1249
Interface	GraphVAE	SD	semi	shared static	32	2.6705	0.3185	-0.0230	1.2466
Interface	GraphVAE	SD	semi	static+dynamic	8	2.6743	0.3039	-0.0243	0.8622
Interface	GraphVAE	SD	semi	dynamic only	32	2.6855	0.3076	-0.0322	1.0470
Interface	GraphVAE	SD	semi	static+dynamic	32	2.6974	0.2941	-0.0411	0.9357

Appendix C

Repository Layout

The source code for the study is hosted at `https://github.com/RohanDevereux/Bridging`

The main repository layout is:

- Molecular-dynamics preparation and simulation:
  `https://github.com/RohanDevereux/Bridging/tree/main/src/bridging/MD`
- Residue-graph preparation:
  `https://github.com/RohanDevereux/Bridging/tree/main/src/bridging/graphvae/prep`
- GraphVAE, supervised baseline, and latent-ridge evaluation:
  `https://github.com/RohanDevereux/Bridging/tree/main/src/bridging/graphvae/ml`
- Command-line runners for the final evaluation:
  `https://github.com/RohanDevereux/Bridging/tree/main/src/bridging/graphvae/runners`
- Shared affinity-conversion and dataset utilities:
  `https://github.com/RohanDevereux/Bridging/tree/main/src/bridging/utils`
- Curated datasets used in the main analysis:
  `https://github.com/RohanDevereux/Bridging/tree/main/src/bridging/processedData`
- Final 52-configuration matrix summary:
  `https://github.com/RohanDevereux/Bridging/blob/main/RESULTS/final_resample_matrix_summary.csv`
- Saved subgroup analysis from the raw outer-fold prediction files:
  `https://github.com/RohanDevereux/Bridging/tree/main/RESULTS/final_resample_subgroups`

Appendix C. Repository Layout

- Subgroup-export script:
  `https://github.com/RohanDevereux/Bridging/blob/main/src/bridging/graphvae/tools/export_final_resample_subgroups.py`
- Archived branches discussed in Appendix A:
  `https://github.com/RohanDevereux/Bridging/tree/main/ARCHIVE`

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